7.1 Seizures

Seizures and status epilepticus requires prompt evaluation and treatment. The likely cause of the seizure including brain lesions, medication non-adherence, overdose, toxic exposure, or metabolic disorders should be used to guide therapy.

NOTE: **Seizures due to eclampsia should be managed as per the** eclampsia guidelines.
NOTE: **Check a bedside blood glucose and** treat hypoglycemia **straightaway.**

- Administer one of the following benzodiazepines every 2–3 minutes until the seizures are controlled.
- IV administration is preferred for treating seizures. Lorazepam IV is preferred, but midazolam IV is an acceptable alternative.

Lorazepam

Adult

- 4 mg IV, IM, NAS

Pediatric

- 0.1 mg/kg IV, IM, NAS
- Max single dose 4 mg

NOTE: **Preferred benzodiazapine if there is IV or IO access.**
NOTE: **Repeat every 2–3 minutes until the seizure is controlled.**

Midazolam

Adult

- 0.1 mg/kg IV
- Max single dose 5 mg

Alternately

- 0.2 mg/kg NAS, IM
- Max single dose 10 mg

Pediatric

- Refer to adult dosing

NOTE: **Preferred benzodiazapine if IM or NAS administration is indicated.**
NOTE: **Repeat every 2–3 minutes until the seizure is controlled.**

⚠ CAUTION **Neuromuscular blocking agents can mask the motor manifestations of seizures and ongoing status epilepticus; however, they do not treat the underlying condition. These agents should not be used as a treatment for status epilepticus.**

7.1.1 Refractory Seizures and Ongoing Seizure Control

One of the following anticonvulsants may be administered for status epilepticus, recurrent seizures, or to maintain seizure control.

Choose one (in order of preference)

NOTICE **The use of propofol, PHENobarbital, or benzodiazepine infusions to treat status epilepticus should be done under direct medical control.**

Levetiracetam—Seizures, Status Epilepticus

Adult

- 4 gm IV
- Administer over 15 min

Pediatric

- 60 mg/kg IV
- Administer over 15 min
- Max single dose 4 gm

Fosphenytoin

Adult

- 15 mg PE/kg IV
- Administer no > 150 mg PE/min

NOTE: **Administer via infusion pump**

Pediatric

- 15 mg PE/kg IV
- Administer no > 3 mg PE/kg/min (max 150 mg PE/min)

Phenytoin

Adult

- 15 mg/kg IV
- Administer no faster than 50 mg/min

NOTE: **Administer via infusion pump**

Pediatric

- 15 mg/kg IV
- Max single dose 1 gm
- Administer no faster than 1mg/kg/min (max 50 mg/mi)

7.2 Acute Ischemic Stroke

A well-founded diagnosis of Acute Ischemic Stroke (AIS) should be made by the attending physician prior to Medical Crews implementing therapy in this section.

7.2.1 Neurological Exam

Perform and document a neurological exam

- Determine time of symptom onset or last time the patient was known to be normal
- Check a bedside blood glucose and treat hypoglycemia straightaway
- Perform a neurological exam every 15 minutes and report any significant changes

7.2.2 Thrombolytic Therapy

- If the patient is receiving or will receive thrombolytic therapy for AIS, review eligibility requirements, contraindications and dosing.

Thrombolytic Therapy Inclusion Criteria

- Clinical diagnosis of ischemic stroke causing neurological deficit.
- Treatment within 4.5 hours of symptom onset.
- Age ≥18 years.

Thrombolytic Therapy Exclusion Criteria

Head CT scan
• Evidence of hemorrhage
History
• Previous intracranial hemorrhage • Ischemic stroke or severe head trauma in the previous three months • Intracranial neoplasm • Gastrointestinal hemorrhage in the previous 21 days • Intracranial or intraspinal surgery within the prior three months
Assessment
• Symptoms suggestive of subarachnoid hemorrhage • Persistent blood pressure elevation (systolic ≥185 mmHg or diastolic ≥105 mmHg) • Active internal bleeding • Stroke associated with aortic arch dissection • Acute bleeding diathesis
Hematology
• Platelet count <100,000/mm3* • Current anticoagulant use with an INR >1.7 or PT >15 seconds or aPTT >40 seconds

Alteplase—Acute Ischemic Stroke (AIS)

Adult

- 0.9mg/kg total dose up to 90 mg maximum
- 10% of the 0.9mg/kg TBW total dose IV over 1 minute
- 90% of the 0.9mg/kg TBW total dose over 60 min via infusion pump
- Requires an online physician order to initiate

NOTE: **Dose calculation should be based on total body weight**

NOTE: **Medical crews may continue existing infusion using profile guidance**

Pediatric

- No PHI Standing Orders or Indications

7.2.3 AIS Blood Pressure Management

NOTICE

Acute elevation in blood pressure may be an appropriate response to maintain brain perfusion. Treating hypertension is a balance between the risk of severe BP surges, and the impact on neurologic function when blood pressure is lowered.

If practicable, medical crews should manage blood pressure collaboratively with treating providers or program medical directors. If medical direction is unavailable, medical crews should treat hypertension under the following conditions and guidelines:

Patients Who are Not a Candidate for Thrombolytic Therapy or Intervention

For patients who are not treated with thrombolytic therapy or mechanical thrombectomy, blood pressure should be treated acutely only when the **systolic blood pressure is >220 mmHg or the diastolic blood pressure is >120 mmHg.**

NOTICE

During the first 24 hours after onset of stroke, limit BP reduction to 15% of the baseline BP. Include prior to arrival treatment and BP reduction when determining the target BP during transport.

BP Reduction to Qualify a Patient for Thrombolytic Therapy

Before thrombolytic therapy is started, the systolic blood pressure should be reduced and then **maintained at or below 180/105 mmHg**

Patients Treated with Thrombolytics

For patients who receive thrombolytic treatment, the blood pressure should be **maintained at or below 180/105 mmHg** for at least 24 hours after treatment.

Patients Treated with Mechanical Thrombecotmy

The use of antihypertensives and BP management before and after reperfusion with mechanical thrombectomy varies across interventional centers. Blood pressure management for pre and post clot retrieval should be at the direction of the interventionalist.

CAUTION

Contact medical control if there is evidence of Intracranial hemorrhage (ICH) or sudden neurological deterioration occurs in patients treated with alteplase.

AIS BP Management Guide

Treatment Status	Systolic	Diastolic	Treatment Authority	Comment
No TPA or Intervention	Treat if > 220 mmHg	Treat if > 120 mmHg	Standing order	Limit BP reduction to 15% of baseline in first 24 hr
To Qualify for TPA	Reduce and maintain ≤ 180 mmHg	Reduce and maintain ≤ 105 mmHg	Standing order	
Post TPA	Maintain ≤ 180 mmHg	Maintain ≤ 105 mmHg	Standing order	
Post thrombectomy	At the direction of the interventionalist		On line order	

7.2.4 Antihypertensives

Choose one medication for blood pressure reduction

Nicardipine

Adult

- 5 mg/hr IV infusion
- Increase by 2.5 mg/hr every 5–15 minutes PRN
- Titrate to BP target
- Max 15 mg/hr

NOTE: **The dose may be reduced to 3 mg/hr after targeted response is achieved**

Ped

- No PHI Standing Orders or Indications

Clevidipine

Adult

- Initiate infusion at 1-2 mg/hr
- Dose may initially be doubled every 90 sec
- As BP approaches goal, reduce the rate of increase by less than double and lengthen time between dose adjustments to every 5–10 min.
- Max dose 32 mg/hr

Ped

- No PHI Standing Orders or Indications

Labetalol

Adult

- 20 mg IV over 2 minutes
- Repeat 40 mg in 10-20 min PRN
- Repeat 80 mg every 10-20 min up to 3 times PRN
- Alternately 2mg/min IV infusion
- Administer to target BP. HR, or max total dose of 300 mg

NOTE: **Labetalol is a combined alpha and beta blocking agent**

Pediatric

- No PHI Standing Orders or Indications

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7.3 Hemorrhagic Stroke

A well-founded diagnosis should be made by the attending physician prior to Medical Crews implementing therapy in this section.

- Perform a neurological exam every 15 minutes and report any significant changes
- Specifically assess for home use or recent administration of anticoagulants

NOTE: **The Medical Crews must manage anticoagulant reversal under direct medical control**

Blood Pressure Management

Where practicable, medical crews should manage blood pressure collaboratively with treating providers or medical control. If medical direction is unavailable the medical crews should treat hypertension using the following guidelines:

- Intracranial hemorrhage with SBP > 160 mmHg
- Lower SBP to 130–160 mmHg

Choose one medication for blood pressure reduction:

Nicardipine

Adult

- 5 mg/hr IV infusion
- Increase by 2.5 mg/hr every 5–15 minutes PRN
- Titrate to BP target
- Max 15 mg/hr

NOTE: **The dose may be reduced to 3 mg/hr after targeted response is achieved**

Ped

- No PHI Standing Orders or Indications

Clevidipine

Adult

- Initiate infusion at 1-2 mg/hr
- Dose may initially be doubled every 90 sec
- As BP approaches goal, reduce the rate of increase by less than double and lengthen time between dose adjustments to every 5–10 min.
- Max dose 32 mg/hr

Ped

- No PHI Standing Orders or Indications

Labetalol

Adult

- 20 mg IV over 2 minutes
- Repeat 40 mg in 10-20 min PRN
- Repeat 80 mg every 10-20 min up to 3 times PRN
- Alternately 2mg/min IV infusion
- Administer to target BP. HR, or max total dose of 300 mg

NOTE: **Labetalol is a combined alpha and beta blocking agent**

Pediatric

- No PHI Standing Orders or Indications

Esmolol

Adult

- Initial load: 500 mcg/kg IV over 1 min
- Titrate 50 mcg/kg/min every 5 min to target HR or BP
- Repeat the loading dose of 500 mcg/kg before every titration PRN until near target HR
- Max infusion dose 300 mcg/kg/min

Pediatric

- No PHI Standing Orders or Indications

7.4 Undifferentiated Stroke

Patients with stroke symptoms encountered on scene or in facilities may not have had a head CT or other diagnostic tests to determine if their symptoms are due to ischemic or hemorrhage stroke.

7.4.1 Blood Pressure Management

Patients with stroke symptoms that have not yet had a head CT to determine if their symptoms are due to ischemic or hemorrhage stroke may benefit from blood pressure control.

NOTICE **Acute elevation in blood pressure may be an appropriate response to maintain brain perfusion. Treating hypertension is a balance between the risk of severe BP surges, and the impact on neurologic function when blood pressure is lowered.**

If practicable, medical crews should manage blood pressure collaboratively with online medical direction. If medical direction is unavailable, medical crews may treat hypertension under the following conditions and guidelines:

- Positive stroke assessment
- Finger stick blood glucose 60 mg/dL or >
- SBP goal of <185 mmHg

CAUTION **Hypoglycemia can cause neurologic deficits mimicking stroke. It is important to check the blood sugar and rapidly correct low serum glucose.**

Nicardipine

Adult

- 5 mg/hr IV infusion
- Increase by 2.5 mg/hr every 5–15 minutes PRN
- Titrate to BP target
- Max 15 mg/hr

NOTE: **The dose may be reduced to 3 mg/hr after targeted response is achieved**

Ped

- No PHI Standing Orders or Indications

Clevidipine

Adult

- Initiate infusion at 1-2 mg/hr
- Dose may initially be doubled every 90 sec
- As BP approaches goal, reduce the rate of increase by less than double and lengthen time between dose adjustments to every 5–10 min.
- Max dose 32 mg/hr

Ped

- No PHI Standing Orders or Indications

Labetalol

Adult

- 20 mg IV over 2 minutes
- Repeat 40 mg in 10-20 min PRN
- Repeat 80 mg every 10-20 min up to 3 times PRN
- Alternately 2mg/min IV infusion
- Administer to target BP. HR, or max total dose of 300 mg

NOTE: **Labetalol is a combined alpha and beta blocking agent**

Pediatric

- No PHI Standing Orders or Indications

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7.5 Brain Herniation Syndrome

Brain herniation may occur due to expanding mass lesions associated with conditions such as intracranial hemorrhage, stroke, abscess, tumors, and hydrocephalus.

- If practicable, medical crews should manage herniation syndrome collaboratively with treating providers or PHI medical directors to ensure decisions are informed by comprehensive medical insights and tailored to the patient's specific needs.
- In cases where medical direction is unavailable, medical crews should follow these guidelines to treat herniation syndrome.

7.5.1 Signs of Brain Herniation Syndrome

- **Neurogenic Breathing Patterns:** Monitor for patterns including Cheyne-Stokes, hyperventilation, and ataxic breathing.
- **Abnormal Pupillary Responses:** Look for abnormalities such as asymmetrical dilation and fixed pupils.
- **Abnormal Motor Responses:** Assess for signs such as decerebrate posturing (extensor posturing), decorticate posturing (flexor posturing), hemiparesis, hemiplegia, and flaccidity.
- **Cushing Reflex:** This reflex is characterized by bradycardia, respiratory depression, and hypertension.

7.5.2 Brain Herniation Syndrome Treatment

Sodium Chloride 3%

Adult

- 250 mL IV over 5-15 min

Ped

- 3–5 mL/kg IBW IV over 5-15 min

Alternate Treatment

Mannitol

Adult

- 0.5-1 gm/kg IV over 30-60 min
- Requires on-line order

Ped

- Refer to adult dosing

7.6 CNS Infection

If practicable, medical crews should manage CNS infections collaboratively with treating providers or program medical directors. If medical direction is unavailable, medical crews should treat CNS infections under the following conditions and guidelines:

NOTE: **Antibiotics should be administered as early as possible**
NOTE: **Give cefotaxime and either ceftriaxone or ampicillin based on patient age**

Cefotaxime

Adult

50 mg/kg IV over 3-5 minutes

Pediatric

- See adult dosing

Plus one of the following based on patient age

Ceftriaxone

Adult

- 50 mg/kg IV over 3-5 minutes
- Maximum dose 2 gm

NOTE: **Age ≥ 3 months old**

Pediatric

- See adult dosing

Ampicillin

Pediatric

- 100 mg/kg IV over 3-5 minutes

NOTE: **Age < 3 months old**

9.1 Poisoning 9-2
9.1.1 Opioid Overdose 9-2
9.1.2 Cyanide Poisoning 9-3
9.1.3 Tricyclic Antidepressant (TCA) Overdose 9-4
9.2 Envenomation 9-5
9.2.1 Snake Bite (Pit Viper) 9-5

9.1 Poisoning

General supportive treatment using other guidelines should be implemented for all poisonings. This guideline provides further procedures for the care of poisoned or chemically exposed patients.

NOTICE **The Poison Control Center should be contacted for assistance in managing the patient at 800–222–1222**

- Identify the type and amount of substance the patient was exposed to
- Establish the time of exposure
- If available, medication containers or chemical labels should be photographed, or taken to the hospital with the patient if safe

9.1.1 Opioid Overdose

Administer naloxone for opioid induced life threatening respiratory depression or apnea

Naloxone

Adult

- 1-2 mg IV/IM/IN

NOTE: **Full reversal of opioid intoxication/overdose**

Pediatric

- 0.01 mg/kg IV/IM/IN
- Max single dose 1 mg

Naloxone—Infusion

Adult

- Determine the initial dose required to reinstate breathing
- Administer 2/3 of that initial dose every hour
- Stop infusion if withdrawal signs or symptoms develop

NOTE: **If called upon to manage a naloxone infusion to treat severe opioid intoxication, defer to the sending or receiving physician with respect to dose and titration. In the absence of such guidance use the dosing in this profile.**

Pediatric

- Refer to adult dosing

Naloxone—Newborn Respiratory Depression

Neonatal

- 0.1 mg/kg IV/IM/IN

NOTE: **Respiratory depression refractory to PPV with a history of maternal opioid administration within 4 hours**

9.1.2 Cyanide Poisoning

Indications

- History of smoke inhalation or removal from fire along with any neurological, cardiac or respiratory complaint.

Hydroxocobalamin

Adult

- 5 gm IV infusion over 15 min
- May repeat 1 time
- Max total dose 10 gm

Pediatric

- 70 mg/kg IV infusion over 15 min
- Max single dose 5 gm
- Max total dose 10 gm

NOTE: **May give undiluted IV push in cardiac arrest**

NOTE: **History of smoke inhalation, removal from fire, Ingestion, inhalation, or dermal exposure to cyanide combined with neurological, cardiac or respiratory complaint.**

HydroxocobalaminAdvisories

- Administration of the second dose may range from 15 minutes to 2 hours based on patient condition
- The recommended diluent is 0.9% Sodium Chloride
- Use the diluent included in the kit or follow kit instructions if a diluent is not included
- Use a separate intravenous line for administration due to multiple incompatibilities

9.1.3 Tricyclic Antidepressant (TCA) Overdose

Treatment is indicated if any on the following symptoms occur:

- Cardiovascular instability
- Hypotension
- Rapid CNS deterioration (seizures, coma).
- Prolonged QT interval
- Widened QRS
- Ventricular dysrhythmias

NOTE: Arrhythmias secondary to TCA should be treated first with sodium bicarbonate

Sodium Bicarbonate

Adult

- 50–100 mEq IV over 5–10 min

Pediatric

- 1–2 mEq /kg IV over 5–10 min
- Max single dose 100 mEq

NOTE: May give 1 dose—additional doses require a physician order

Cross reference		
Clinical Care Guidelines	23	7-1 - Neurocritical Care

9.2 Envenomation

- Fully expose and examine the site of the bite or sting
- Observe for edema, redness, and number of puncture sites
- Mark border of swelling with ink and time
- If practicable, immobilize affected extremities in a position that is lower than the patient's heart

⚠ CAUTION **The insect/arachnid/snake should not be brought along on the transport.**

9.2.1 Snake Bite (Pit Viper)

- Measure the circumference of the bitten extremity just proximal to envenomation site every 15 minutes.
- Administer CroFab (Crotalidae Polyvalent Immune Fab [Ovine])

Crotalidae Polyvalent Immune Fab

Adult

- 4–6 reconstituted vials in 250 mL of 0.9% NS IV
- 25–50 mL/hr for 10 minutes observing for allergic reaction
- If no reaction occurs, increase rate to 250 mL/hr

Ped

- No PHI Standing Orders or Indications

Scorpion Sting

Observe for signs of envenomation neurotoxicity

- Flushing
- Diaphoresis
- Agitation
- Salivation
- Roving eye movements (Opsoclonus)
- Limb jerking (these are myoclonic jerking NOT seizures)

Treatment

Secretions

- Manage secretions using suction
- Atropine may be given onetime for hypersalivation

⚠ CAUTION **In addition to standard indications, early advanced airway management may be needed if secretions cannot be managed with suction or atropine.**

Atropine

Adult

- 1 mg IV over 1 min, IM

NOTE: **May be given one time**

Pediatric

- 0.02 mg/kg IV 1 min, IM
- Max single dose 0.5 mg

Sedation and Clonus

- Fentanyl (preferred) or midazolam may be given for clonus or sedation.

NOTE: **Avoid long-acting opioids or benzodiazepines such as hydromorphone, morphine lorazepam or diazepam**
NOTE: **Avoid anticholinergics such as diphenhydramine**

Fentanyl

Adult

- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN

Pediatric

- Refer to adult dosing

Midazolam

Adult

- 0.1 mg/kg IV
- Max single dose 5 mg

Alternately

- 0.2 mg/kg NAS, IM
- Max single dose 10 mg

NOTE: **Preferred benzodiazapine if IM or NAS administration is indicated.**
NOTE: **Repeat every 2–3 minutes until the seizure is controlled.**

Pediatric

- Refer to adult dosing